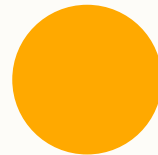




Other Careers in Comp. Sci.

Reese Hatfield



0



Careers

- Most people aim for something software development related
- Not a universal truth
- There are many career options that are still computer science-y
 - (some still program)
- That aren't software development





Careers

- **Data Scientist**
- Data becomes more important every day
- Scale of data grows
- Need someone to analyze that data





Careers

- Companies need to make decisions
- Based upon all their data
- Likely user data
- Data Scientists are tasked with interpreting all of that data to help the company





Careers

- Data scientists are often writing smaller bits of code (scripts, usually python) to help with this task
- Often focused around visualizing patterns
- Identifying outliers, etc.





Careers

- **Cybersecurity Analyst**
- Companies have very large codebases
- On the order of 10's of millions of lines
- Bugs introduce vulnerabilities





Careers

- Adding to this, code is often dependent upon hundreds of other code bases
- Software libraries
- Dependency vulnerabilities
- Let's peek at dependabot *





Careers

- Someone has to be in charge of managing + reporting these vulnerabilities
- **Common Vulnerabilities and Exposures**
- Catalog of common software cybersecurity issues





Careers

- These are only for public facing codebases
- Known vulnerabilities
- Companies don't exactly want them known in the first place
- Internally, a cybersecurity analyst is tasked with addressing this





Careers

- Cybersecurity analysts are also responsible for internal networking monitoring
- Ensure no suspicious activity is occurring on the internal network
- Potentially the external network as well





Careers

- Cybersecurity has many more specialized roles
- Pen Testing is one of the jobs a lot of cyber-focused people go for?
- This is basically an ethical-hacking style job





Careers

- Companies want to ensure they have a secure system
- Effectively paying people to find potential vulnerabilities
- Break into their own systems, before someone else can





Careers

- If this interests you, check the National Cyber League
- Offer competitive cyber security challenges
- "Capture the flag"
- Anyone done these?





Careers

- **Developer Operations Engineer**
- Teams have a lot of developers
- At some point, someone has to help them automate code management
- Usually at the VCS level





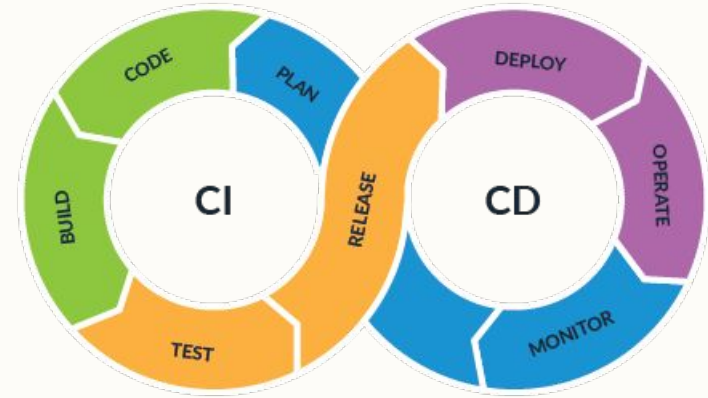
Careers

- **Dev Ops** usually occurs on the commit/push level
- "Continuous integration" pipeline
- This is almost its own entire codebases
- To help ensure developers are functioning correctly



Careers

- Often also responsible for deploying the software
- Code → Released Product





Careers

- **Database Administrator**
- Someone has to design those database schemas that we had previously talked about
- They also need to allow them to scale
- Appropriate usage, etc.





Careers

- Someone needs to administrate that data
- Requirements are constantly shifting
- Migrate databases across different changes
- Think about Amazons DB might have changed throughout the years





Careers

- **UI/UX Engineer**
- Someone has to design the user-facing application
- Think "web design" type of work
- A lot more goes into this than you might expect





Careers

- User interface
- What the user sees
- User eXperience
- How the user feels using the application



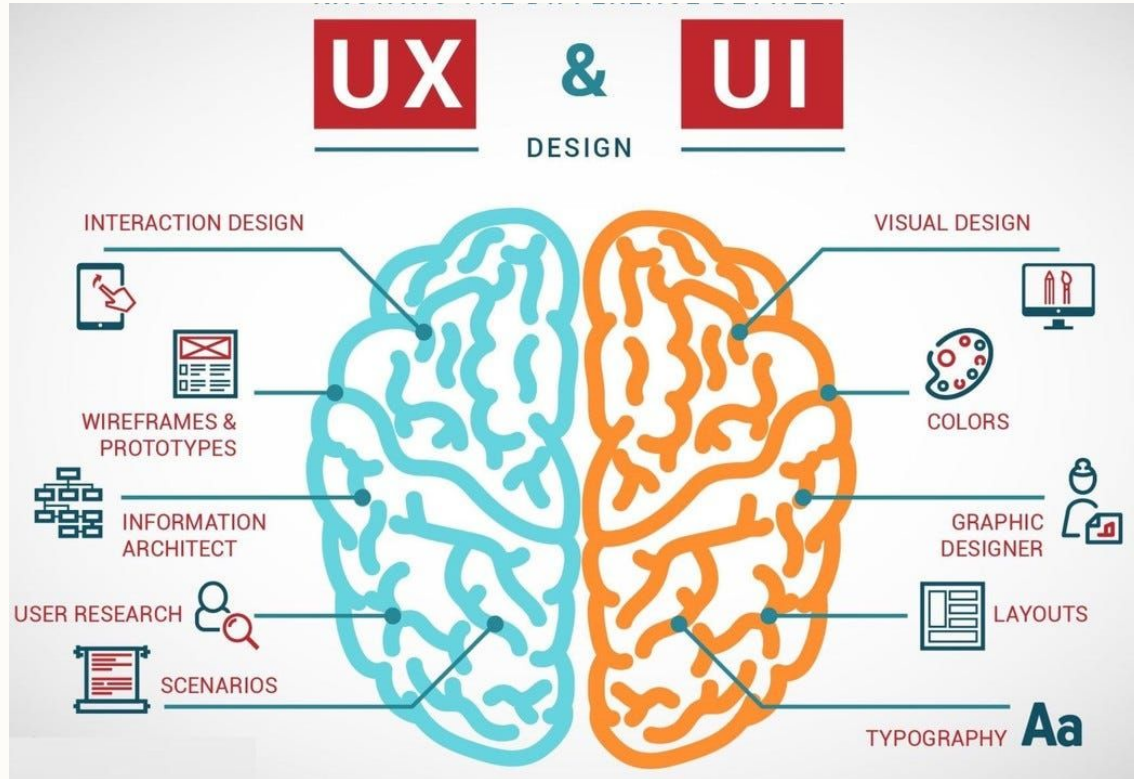


Careers

- User interface
- What the user sees
- User eXperience
- How the user feels using the application



Careers





Careers

- CS careers are not limited to what we have discussed
- Anyone have any other career prospects?
- That's all

